

Backend Engineer (Junior) — Take Home Assignment

Objective

Build a backend service that powers a modular conversation flow system.

This assignment evaluates:

- API design
- Data modeling
- Handling of user state vs conversation history
- Defensive backend design for edge cases
- Ability to use AI tools responsibly and transparently

You are allowed (and encouraged) to use AI tools. However, you must clearly document how you used them (details below).

Problem Statement

You are building a system that drives a question-based conversation flow for users.

The system consists of multiple modules. Each module contains a set of questions.

A user moves through questions by selecting options. Each option determines which question comes next.

During a conversation, a user may:

- Move within the same module
- Switch to a different module
- Re-enter the flow from an old link or notification
- Reach special questions that reset previous context for that module

Your task is to design and implement the backend that correctly handles all of the above.

You are free to design the APIs, data structures, and storage as you see fit.

Functional Expectations

Your system must support the following behaviors:

1. A user can start a module and begin answering questions.
 2. Each question presents options that determine the next question.
 3. Some options may move the user to a different module.
 4. The system must maintain:
 - Complete conversation history for the user
 - Active state of the user within the current module
 5. Some questions act as checkpoints:
 - After such a question, previous context for that module should not affect future flow.
 - History must still be preserved.
 6. The system must handle a user opening an old deep link or notification pointing to a specific question:
 - If that question is no longer valid, the system should return the latest valid question based on the user's current state.
 7. The system must defensively handle incorrect flows such as:
 - Invalid options
 - Broken question references
 - Switching between modules multiple times
 - Returning to a previously visited module
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AI Usage (Optional)

Create a file named: AI_USAGE.md

This file must include:

1. AI tools you used
2. Prompts you gave
3. What you modified from AI output
4. What AI got wrong
5. How you verified correctness

We will also evaluate how you think while using AI, not whether you avoid it.

Bonus (Optional)

Allow users to go back to the previous question within the current module state.

Submission

- GitHub repository
 - README with setup instructions
 - `AI_USAGE.md`
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Evaluation Criteria

We will evaluate:

- Quality of API and data design
- Code clarity and structure
- Handling of state vs history
- Defensive handling of edge cases
- Thoughtful and transparent AI usage